

Ex No 2

30/11/26

Study of Network Cables

1. understand different types of Network cables.

AIM: understand different types of network cable.

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable.
2. Shielded Twisted Pair (STP) cable.
3. Coaxial cable
4. Fibre optic cable.

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use	Image
UTP	Category 3	10 bps	<u>Advantages</u> • Cheaper in Cost • Easy to install as they have a smaller overall diameter.	10 Base-T Ethernet	
	Category 5	up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10 Gbps	<u>Advantages</u> • Shielded • Faster than UTP. • Less susceptible to noise and interference.	Gigabit Ethernet, 10 G Ethernet (55 m) widely used in data centers.	  

Cable type	Category	Maximum Data Transmission	Advantages/Disadvantages	Application / use	Image
SSTP	Category 7	10 Gbps	<u>Disadvantages</u> • Expensive • Greater installation effort • High bandwidth • Immune to interference	Gigabit Ethernet, 10 G Ethernet (100 m)	
Coxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	• Low loss bandwidth • Versatile <u>Disadvantages</u> • Limited distance • Cost • size is bulky.	Speed of signal is 500m Television network High speed internet connections	
Fibre optics Cable	Single mode Multi mode	100 Gbps	<u>Advantages</u> • High speed • High bandwidth • High security • Long distance <u>Disadvantages</u> • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100 meters	

Result:

The types of network cables and making cross wired cable straight through cable using clamping / crimping tool is studied.